

Saber Ayatollahi, CFA, MSc

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SUMMARY

Quantitative risk professional with ~7 years across VaR/CVaR analytics, derivatives pricing validation, and Monte Carlo scenario modelling, applying directly to the Associate Director, Global Risk Analytics mandate at RBC. Holds delegated sign-off on valuation and sensitivity outputs for multi-asset institutional portfolios (\$5-25bn per engagement) at Moody's Analytics; built stochastic scenario generators and ran VaR/CVaR portfolio optimization at Ortec Finance. Python, SQL, MATLAB (research background); CFA charterholder with dual MSc degrees (Financial Modelling, Chemical Engineering).

CORE SKILLS

Market Risk Analytics & Measurement • VaR & CVaR (Expected Shortfall) • Stress Testing & Scenario Analysis • Derivatives Pricing Validation (Rates, FX, Inflation) • Fixed Income & Interest Rate Analytics • Monte Carlo & Stochastic Scenario Generation • Yield Curve Construction & Calibration • Model Governance, Documentation & Sign-Off • Python (pandas, NumPy, SciPy) - SQL - MATLAB (research background)

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services

Toronto, May 2023 - Present

Methodology Review, Documentation & Model Governance

- Holds delegated sign-off authority on valuation, sensitivity, and risk outputs for multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing yield curve construction, spread calibration, and cross-asset interactions prior to production release.
- Serves on the model governance committee, contributing to methodology review, benchmarking and documentation standards, and model-performance assessment for client-delivered analytics.
- Escalates outputs lacking mathematical or economic defensibility; acts as escalation point for client-critical analytical issues with clients' Heads of Risk and investment stakeholders.

Market Risk Methodology Review & Derivatives Pricing Validation

- Validated derivatives pricing outputs across interest rate, foreign exchange, and inflation products, confirming sensitivity and scenario-impact consistency at portfolio-level aggregates.
- Analyzed interest rate risk and duration exposure under parallel and non-parallel rate shocks across fixed income and derivative positions.
- Reviewed aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream risk and capital processes.

Model Implementation & Scenario Engineering

- Led design and implementation of an enterprise multi-asset cash flow and stress-scenario engine supporting base, stress, and reverse-stress runs across T+1 to multi-year horizons.
- Re-engineered manual spreadsheet workflows into scalable Python (pandas, NumPy, SciPy) analytics pipelines with embedded logging, validation, and auditability controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and documentation, cutting development cycle time an estimated 30-40% on comparable modules.
- Prepared analytical summaries for senior stakeholders on market exposures, scenario impacts, and portfolio sensitivities, translating model output into committee-ready narratives.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds and asset managers migrating onto the PFaroe DB (risk & ALM analytics) and PFaroe PM (portfolio management) platforms; scoped investment-team requirements into development requests for product owners and validated model outputs post-deployment.

Ernst & Young (EY)**Senior Consultant - IFRS & Financial Risk**

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 technology-transformation projects for Canadian and international insurance clients, aligning regulatory requirements with system implementation.
- Coordinated across actuarial, finance, IT, and risk functions on milestone delivery, governance documentation, and regulatory-readiness processes.

Ortec Finance Canada**Consultant - ALM & Risk**

Toronto, Feb 2019 - Sep 2021

- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition, contribution-to-risk budgeting, and near-optimal frontier analysis to test allocation robustness.
- Built and interpreted stochastic (Monte Carlo) economic scenario generators assessing interest rate risk and funding volatility under base-case and stressed environments.
- Conducted cash flow, liquidity, and hedging analysis across interest rate, inflation, and currency exposures for international institutional clients.
- Delivered ALM studies for Canadian and international pension funds, analyzing funding ratios, duration mismatch, and long-horizon balance-sheet risk across all major asset types.
- Presented study findings, stress testing results, and allocation recommendations to investment committees at Canadian pension-fund clients on-site.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016